

NACL (NETWORK ACCESS CONTROL LIST)

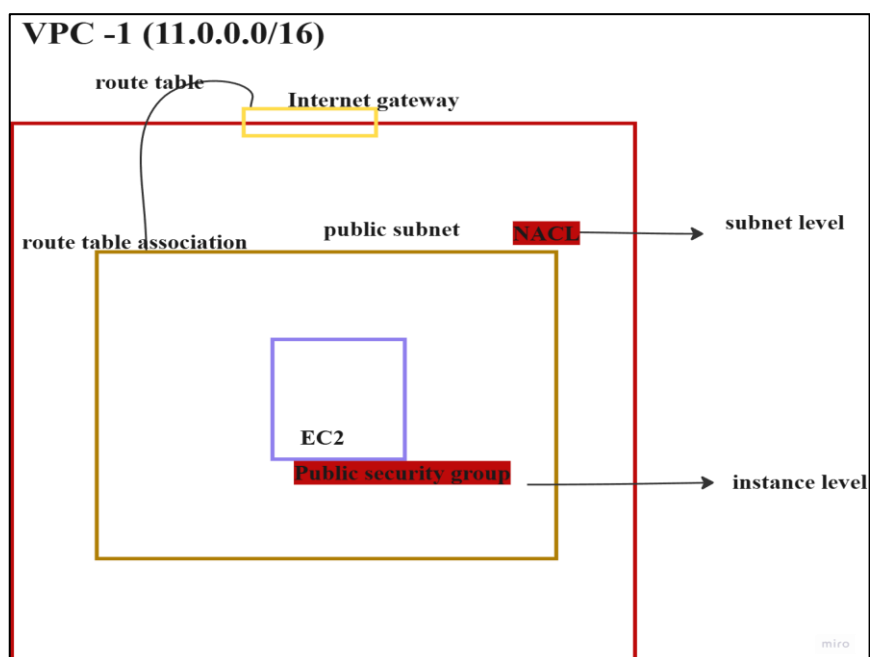
A Network Access Control List (NACL) is a stateless virtual firewall that operates at the subnet level in a VPC, controlling inbound and outbound traffic by using a set of ordered allow and deny rules.

SECURITY GROUP

A Security Group in Amazon Web Services is a stateful virtual firewall that controls inbound and outbound traffic to resources such as EC2 instances by allowing only specified traffic based on defined rules.

SECURITY GROUP VS NACL

Feature	Security Group	NACL (Network ACL)
Level	Works at instance level	Works at subnet level
Rules	Allow only rules	Allow & Deny rules
State	Stateful (remembers traffic)	Stateless (no memory)
Traffic Check	Checks mainly inbound (auto allows return)	Checks inbound & outbound separately
Rule Processing	All rules evaluated together	Rules processed in number order



HANDS-ON FOR NETWORK ACL

Screenshot 1: Launch EC2 Instance

- Go to Amazon Web Services Console
- Navigate to EC2 Dashboard
- Click Launch Instance
- Enter instance name
- Select Amazon Linux AMI
- Choose/Create Key Pair
- Under Network settings, allow:
 - SSH (port 22)
 - HTTP (port 80)
 - HTTPS (port 443)
- Click Launch Instance

aws [Search] [Alt+S] Asia Pacific (Mumbai) Jai Surya (2018-0461-1986) Jai Surya

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices. [Take a walkthrough](#) [Do not show me this message again.](#)

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents | My AMIs | **Quick Start**

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

▼ Summary

Number of instances [Info](#)
1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...[read more](#)
ami-0f559c3642608c138

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

[Create new key pair](#)

Firewall (security groups) Info
 A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-5' with the following rules:

- ☒ Allow SSH traffic from Helps you connect to your instance. Anywhere (0.0.0.0/0)
- ☒ Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server.
- ☒ Allow HTTP traffic from the internet To set up an endpoint, for example when creating a web server.

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage Info Advanced

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

Summary

Number of instances Info: 1

Software Image (AMI)
 Amazon Linux 2023 AMI 2023.10...[read more](#)
 ami-0f559c3642608c138

Virtual server type (instance type)
 t3.micro

Firewall (security group)
 New security group

Storage (volumes)
 1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Screenshot 2: Connect to Linux Instance

- Go to **Instances**
- Select your Linux instance
- Click **Connect**

Instances (1/1) Info

[Find Instance by attribute or tag \(case-sensitive\)](#) All states

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
☒	linux	i-0e9974ead13338dd	Running	t3.micro	Initializing	View alarms +	ap-south-1a	ec2-43-205-112-129.ap...	43.205

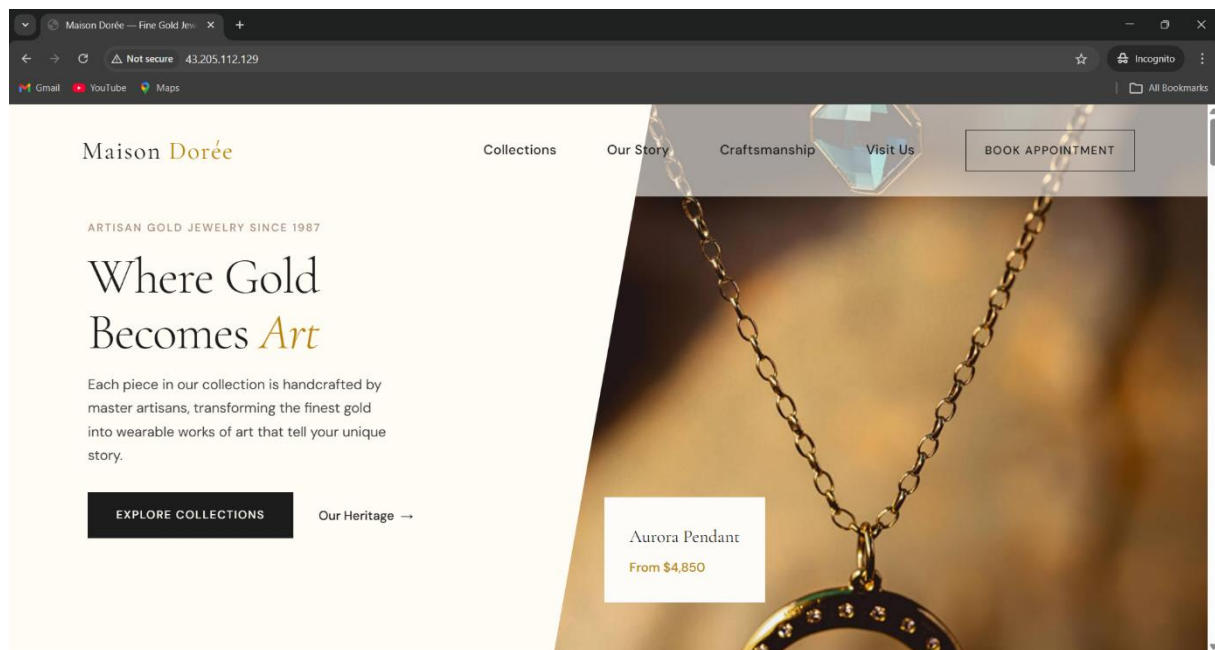
Screenshot 3: Execute HTTPD Web Hosting Commands

- Execute HTTPD web hosting commands in Linux terminal

```
[root@ip-172-31-34-101 html]#
[root@ip-172-31-34-101 html]# history
1 yum update -y
2 yum install httpd -y
3 systemctl status httpd
4 systemctl start httpd
5 systemctl status httpd
6 wget -S --content-disposition https://templatemo.com/download/templatemo_611_maison_doree
7 unzip templatemo_611_maison_doree.zip
8 ls
9 cd templatemo_611_maison_doree/
10 mv * /var/www/html
11 mv * /var/www/html
12 ls
13 cd /var/www/html
14 ls
15 history
```

Screenshot 4: Website Opened

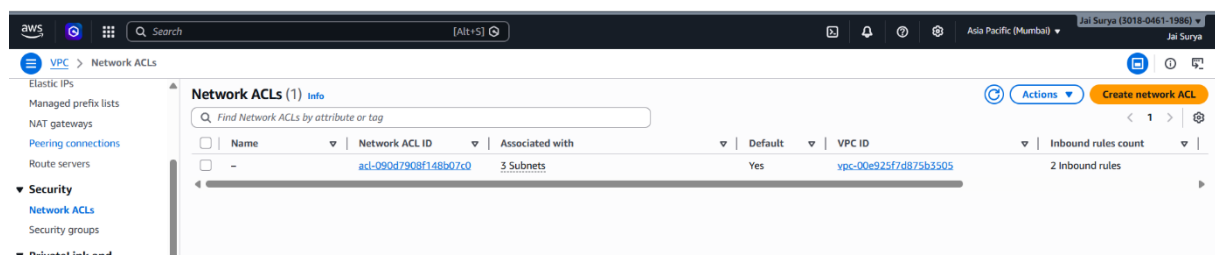
- Open the web browser
- Enter the public IP of the instance
- Verify that the website is displayed successfully



DENY RULE IN NACL

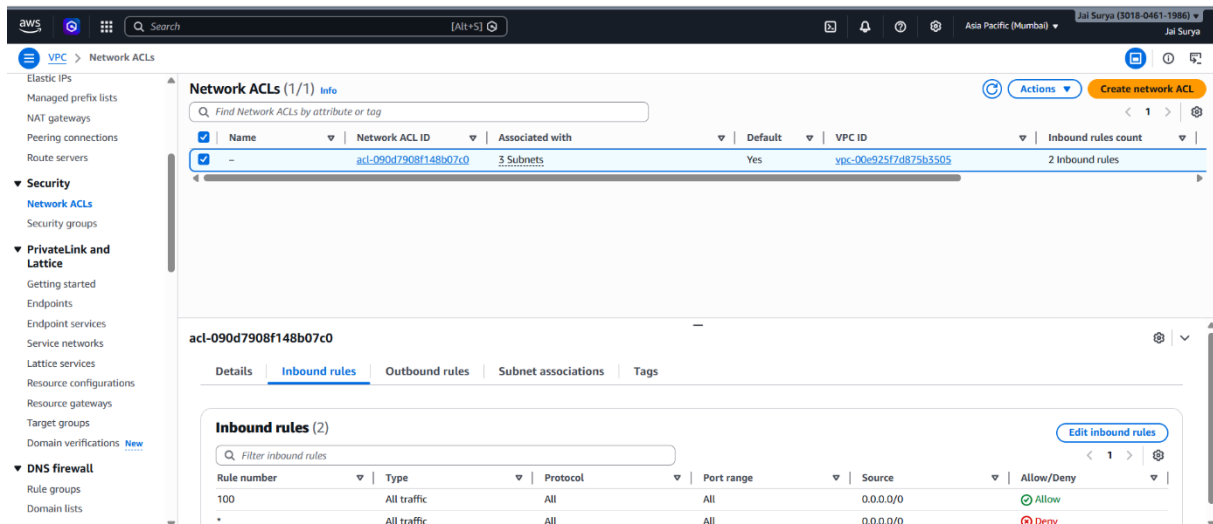
Screenshot 5: Network ACLs

- Go to **VPC**
- Click **Network ACLs**



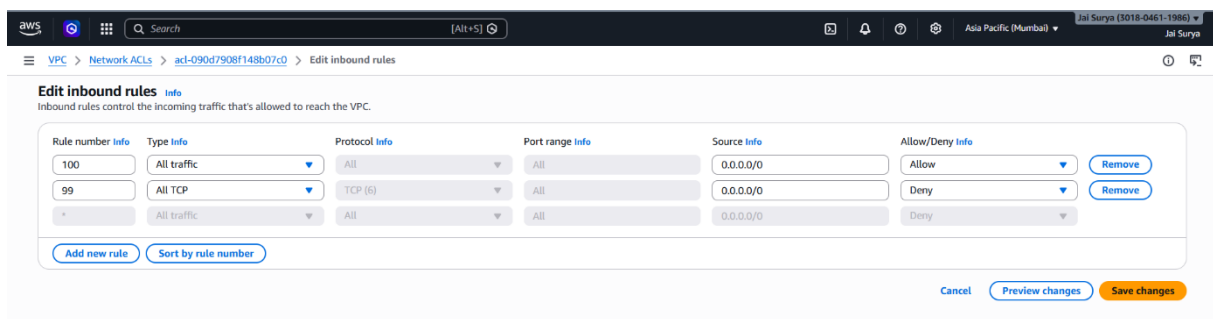
Screenshot 6: Edit NACL Inbound Rules

- Select **Default NACL**
- Go to **Inbound Rules**
- Click **Edit**



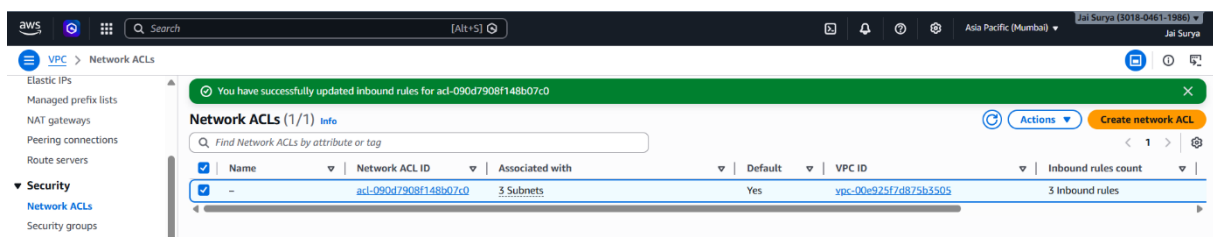
Screenshot 7: Add Deny Rule

- Click **Add Rule**
- Enter rule number **99**
- Select **All TCP**
- Set action to **Deny**
- Click **Save**



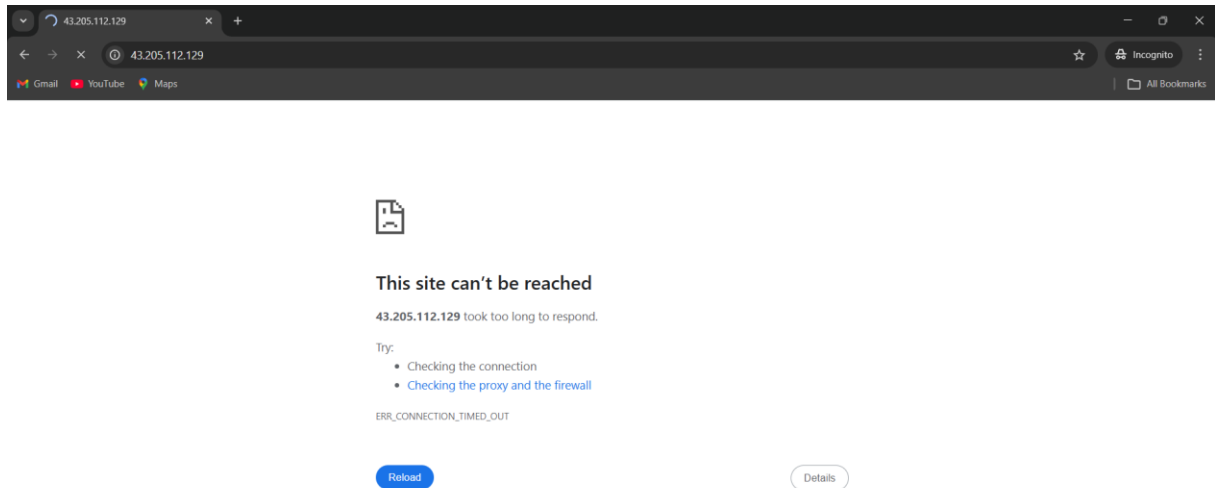
Screenshot 8: NACL Rule Updated Successfully

- Verify that the new rule is added
- Confirm the rule is updated successfully



Screenshot 9: Website Not Reachable

- Go to the web browser
- Refresh the website page
- Verify that the site is not reachable due to NACL deny rule



ELASTIC IP

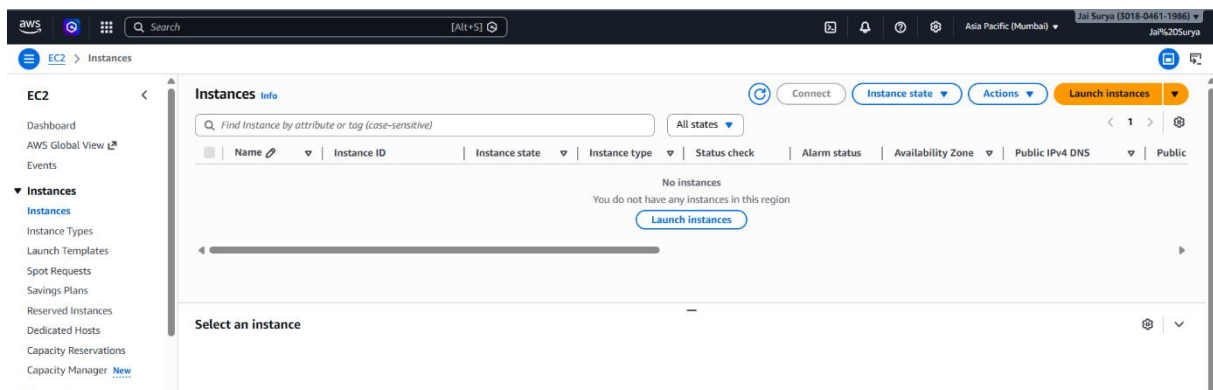
An Elastic IP in **Amazon Web Services** is a **static, public IPv4 address** designed for dynamic cloud computing, which can be associated with an EC2 instance and remains constant even if the instance is stopped or restarted.

HANDS-ON FOR ELASTIC IP

BEFORE USING ELASTIC IP

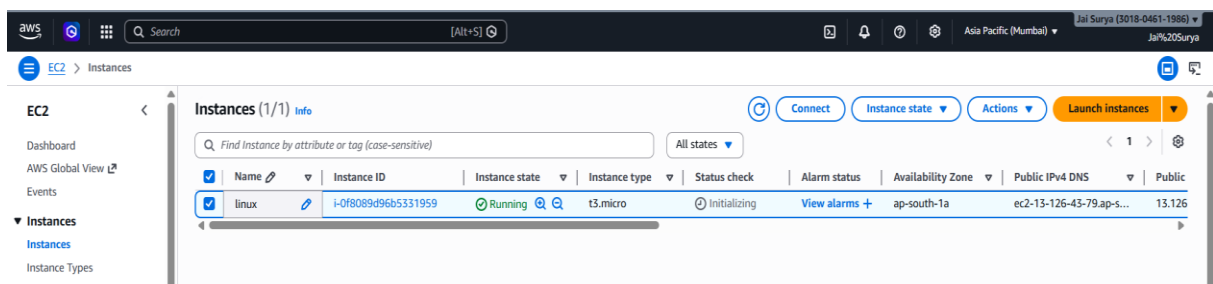
Screenshot 1: Create Linux Instance

- Go to Amazon Web Services Console
- Navigate to EC2
- Click Launch Instance
- Enter instance name
- Select Amazon Linux AMI
- Select key pair
- Click Launch Instance



Screenshot 2: Instance Created

- Go to Instances
- Verify that the instance is created and running



Screenshot 3: Verify Public IPv4 Address

- Go to Instances
- Select your instance
- Check the Public IPv4 address
- Verify the IP is assigned

Instance summary for i-0f8089d96b5331959 (linux) [Info](#)

Updated less than a minute ago

Instance ID

 i-0f8089d96b5331959

IPv6 address

—

Public IPv4 address

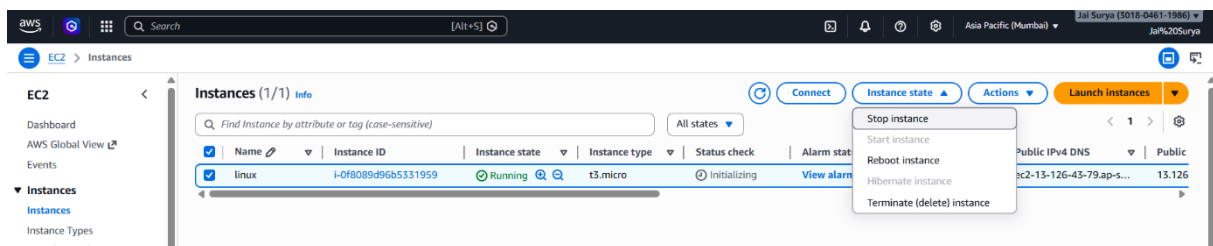
 13.126.43.79 | [open address](#)

Instance state

 Running

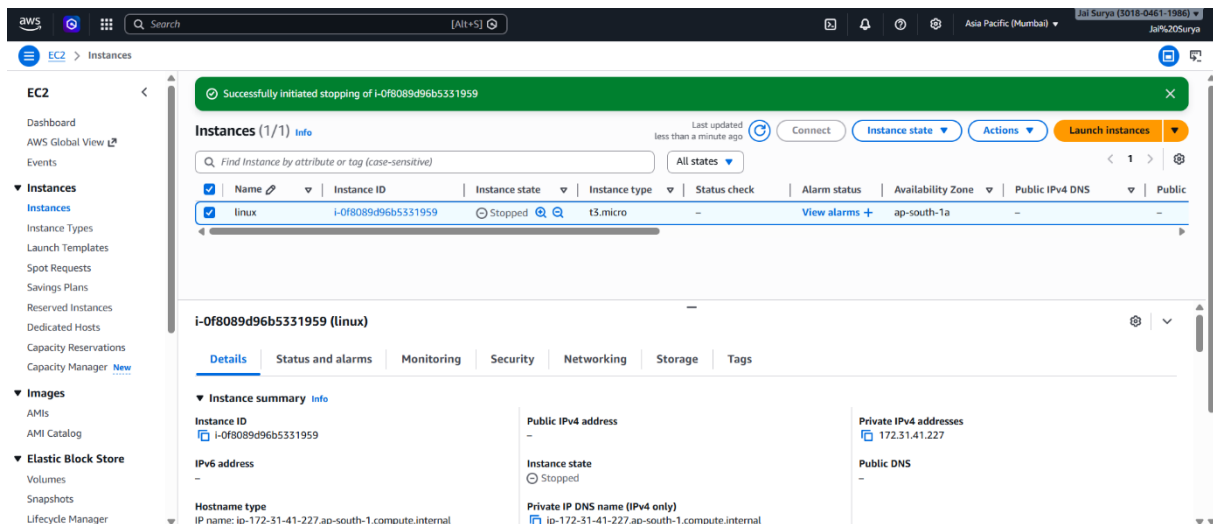
Screenshot 4: Stop Instance

- Select the instance
- Click Actions
- Click Instance State → Stop
- Confirm to stop the instance



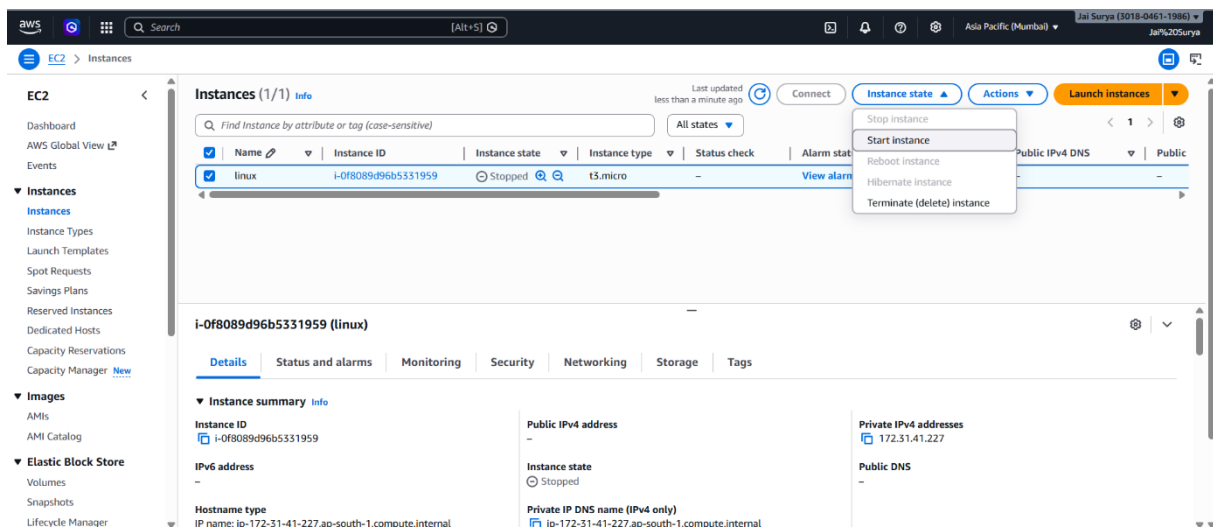
Screenshot 5: Public IPv4 Removed

- Verify that the instance is stopped
- Check the Public IPv4 address
- Observe that the public IP is removed



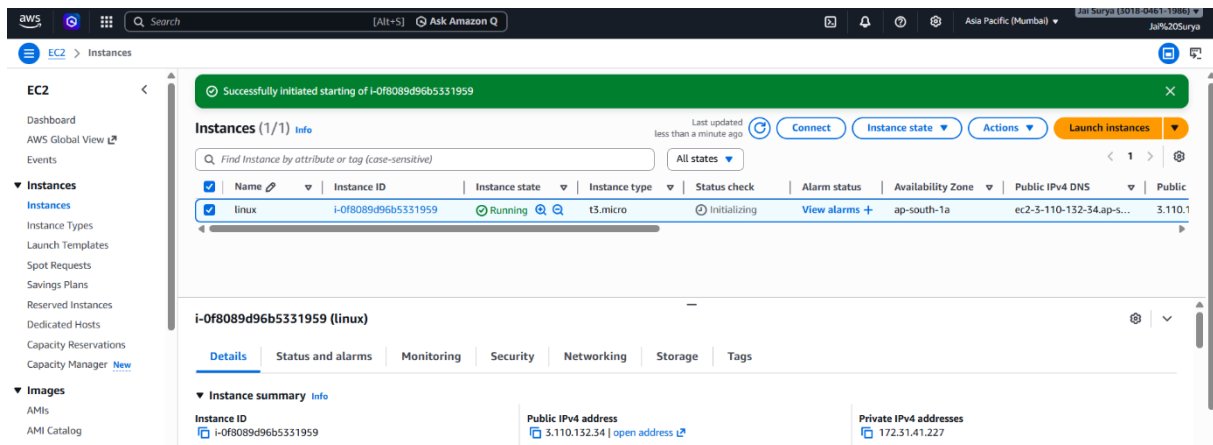
Screenshot 6: Start Instance

- Select the instance
- Click Instance State → Start
- Wait for the instance to run



Screenshot 7: Public IP Changed

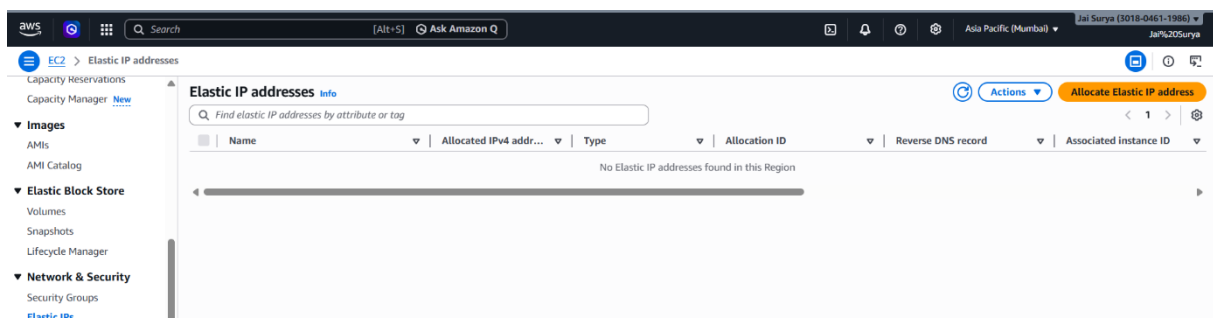
- Verify that the instance is running
- Check the Public IPv4 address
- Observe that the public IP has changed



AFTER USING ELASTIC IP

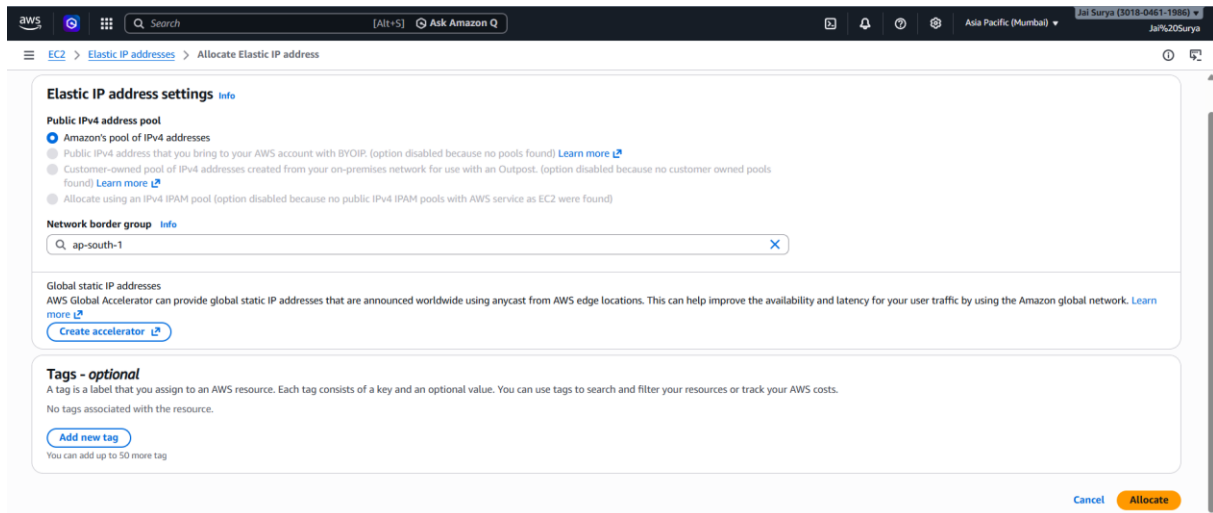
Screenshot 8: Allocate Elastic IP

- Go to Elastic IPs
- Click Allocate Elastic IP address



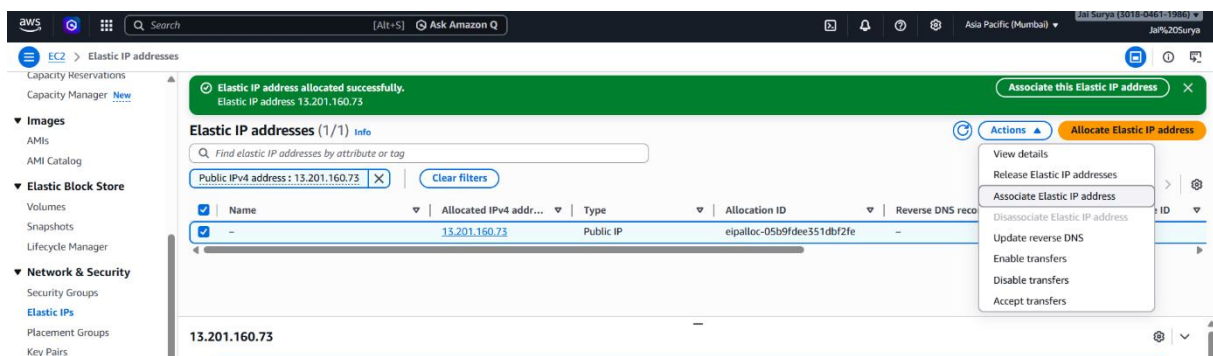
Screenshot 9: Select Network Border Group

- In Elastic IP allocation page
- Select Network Border Group
- Choose the required region
- Click Allocate



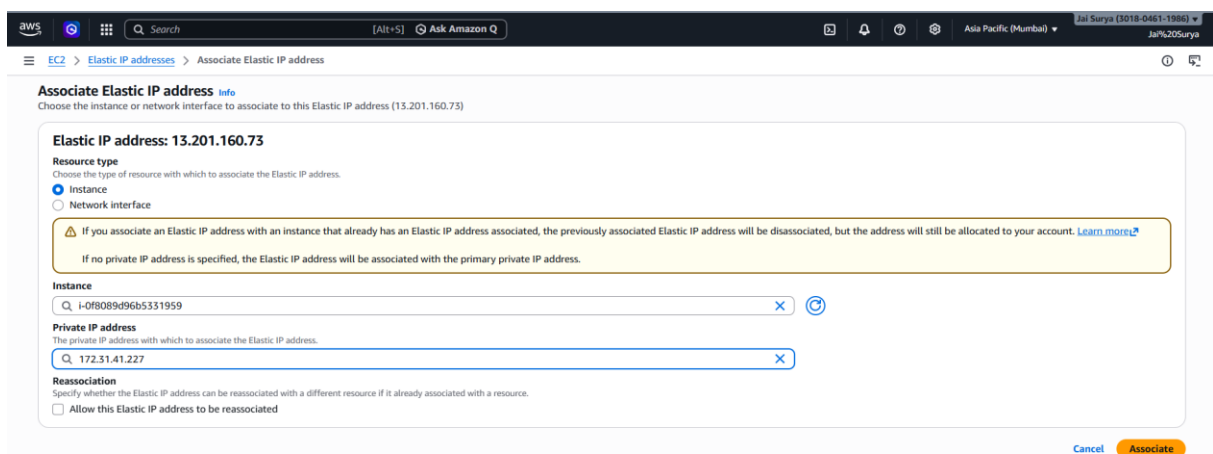
Screenshot 10: Associate Elastic IP

- Select the created Elastic IP
- Click Actions
- Click Associate Elastic IP



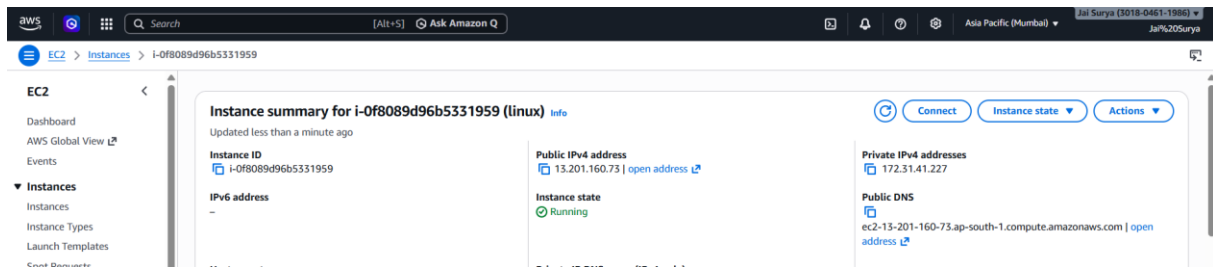
Screenshot 11: Select Instance and Private IP

- Select your instance
- Select the private IP address
- Click Associate / Enter



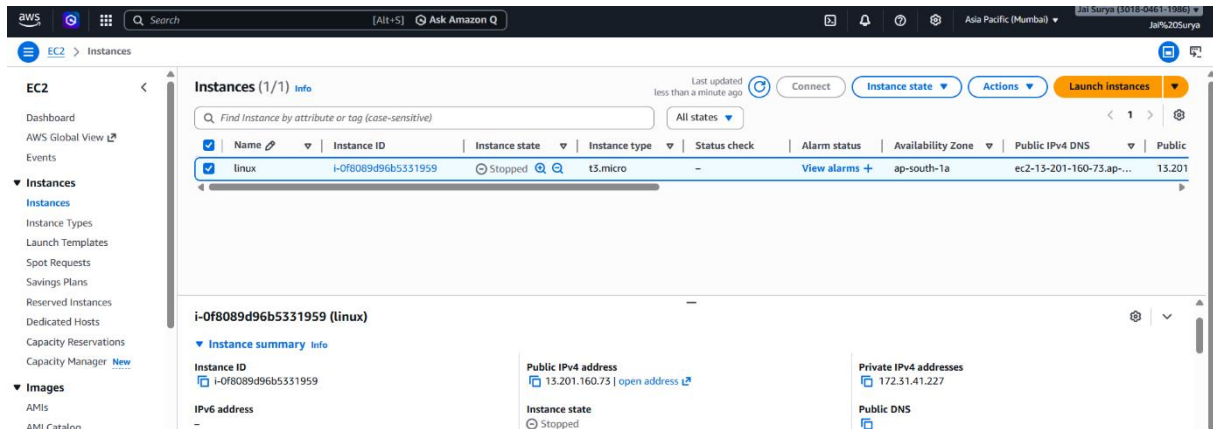
Screenshot 12: Verify Public IPv4 Address

- Go to Instances
- Select your instance
- Check the Public IPv4 address
- Verify that it matches the Elastic IP



Screenshot 13: Stop Instance and Verify IP

- Select the instance
- Click Instance State → Stop
- After stopping, check the Public IPv4 address
- Verify that the Elastic IP is not removed



Screenshot 14: Start Instance and Verify IP

- Select the instance
- Click Instance State → Start
- After starting, check the Public IPv4 address
- Verify that the Elastic IP remains unchanged

The screenshot displays the AWS Management Console interface for the EC2 service. A green notification banner at the top states "Successfully initiated starting of i-0f8089d96b5331959". The left sidebar shows the navigation menu with "Instances" selected. The main content area shows a table of instances with one instance, "linux", in the "Running" state. Below the table, the "Instance summary" for "i-0f8089d96b5331959" is displayed, showing the Public IPv4 address as 13.201.160.73 and the Instance state as "Running".

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public
linux	i-0f8089d96b5331959	Running	t3.micro	Initializing	View alarms +	ap-south-1a	ec2-13-201-160-73.ap-...	13.201

i-0f8089d96b5331959 (linux)

▼ Instance summary Info

Instance ID
i-0f8089d96b5331959

IPv6 address
-

Public IPv4 address
13.201.160.73 | open address

Instance state
Running

Private IPv4 addresses
172.31.41.227

Public DNS
ec2-13-201-160-73.ap-south-1.compute.amazonaws.com | open